Risk Factors for Development of Actinic Keratoses

- **Individual susceptibility** - Older age
- Male gender
  - Fair skin that easily burns and freckles
  - Blond or red hair
- Light-colored eyes
- **Cumulative ultraviolet (UV) radiation exposure**
  - Includes exposure from sunlight and tanning beds
- **Immunosuppression**
- **Prior history of actinic keratoses or other skin cancers**
- **Genetic syndromes**
- Xeroderma pigmentosum
- Albinism
- Rothmund-Thomson syndrome
- Bloom syndrome

Cumulative exposure to UV radiation is the other major risk factor for developing actinic keratoses (AKs). Evidence supporting the role of sun exposure includes the observation that over 80% of AKs occur on habitually sun-exposed areas such as the scalp, face, neck, forearms, and dorsal hands. Factors influencing cumulative UV exposure include age, gender, occupation, recreational activities, and geographic location. The prevalence of AKs increases with age, reflecting greater cumulative UV exposure. Exposure to intense UV radiation during childhood appears to confer the highest risk.

Migration studies in Australia show that British immigrants who moved before age 20 had lower rates of AKs early in life compared to native white Australians but eventually developed similar rates later in life. In contrast, those who migrated after age 20 never reached the same AK incidence as native Australians or early-arriving immigrants.

Etiology and Pathogenesis

Although both genetic and environmental factors contribute to the development of AKs and squamous cell carcinoma (SCC), chronic UV radiation exposure—primarily from sunlight—is the most significant factor. UV radiation promotes AK and SCC development through two key mechanisms:

1. Induction of DNA mutations, particularly in the **p53 tumor suppressor gene**, which, if unrepaired, lead to uncontrolled cellular proliferation and tumor formation.
2. Local immunosuppression, impairing immune surveillance and enabling mutated cells to evade detection and destruction (see Chaps. 109–112).

UV-induced p53 mutations are central to the initiation of AKs and their progression to SCC (see Fig. 113-1; Chap. 112). Repeated UV damage initiates a carcinogenic pathway that progresses from photodamaged skin to AKs and, in some cases, to invasive SCC. AKs represent clonal expansions of genetically altered keratinocytes that have escaped apoptosis and immune monitoring, leading to pre-malignant lesion formation.

Clinical Findings

The typical patient with AKs is an older, fair-skinned individual with light-colored eyes, a tendency to burn and freckle rather than tan, and clinical signs of significant photodamage such as solar elastosis (Fig. 113-2). While most commonly seen in older adults, AKs can occur in younger individuals with substantial lifetime sun exposure.

Approximately 80% of AKs appear on chronically sun-exposed areas, including the face, scalp, neck, forearms, and dorsal hands. Common symptoms include pruritus, burning, stinging, bleeding, and crusting. The classic lesion—often termed the **erythematous AK**—typically presents as a 2- to 6-mm, flat, erythematous, rough, or scaly papule. These lesions are often more easily palpable than visible. Lesion size may vary, with some reaching several centimeters in diameter.

AKs usually arise on a background of **photodamaged skin (dermatoheliosis)**, characterized by: - Solar elastosis
- Dyspigmentation
- Yellowish discoloration
- Ephelides (freckles)
- Telangiectasias
- Skin laxity

In some cases, the number and coalescence of lesions are so extensive that they resemble a widespread rash.

In addition to the classic erythematous type, several clinical subtypes of AK exist (see Table 113-3).